Windows Elevation Showdown





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Agenda

- Windows Elevation History
- Security Tokens Evolution
- EPM
- LAPS
- Usage Scenarios

Windows Elevation History





Windows Elevation History





Windows Elevation History

- Windows 1.0 – 3.11: Single user; All apps run with full privileges



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- Windows NT – Windows XP: Multiuser and basic access control; Most apps still require full privileges.



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- Windows Vista – Introduction Mandatory Integrity Controls (MIC) and User Account Control (UAC)



Mandatory Integrity Levels

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System	system, local system, local service and network service
High	administrators and backup operators
Medium	authenticated standard users
Low	everyone
Untrusted	anonymous

access



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What is Dual Enrollment?

Dual enrollment enables administrators to perform elevated, administrative functions by enrolling both their nonprivileged and privileged credentials on their device.



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- Windows 10 –Release of Windows Hello and later support for dual enrolment*
- Windows 11 (Insider only for now) - Administrator protection mode

Security Tokens Evolution





Security Tokens Evolution

Microsoft Windows xp



Security Tokens Evolution

- Single Access Token

Microsoft Windows xp



Security Tokens Evolution

- Single Access Token

- No Elevation Mechanism

Microsoft Windows xp



Security Tokens Evolution

- Single Access Token

- No Elevation Mechanism

- Admin by default

Microsoft Windows xp



Security Tokens Evolution





Security Tokens Evolution

Windows Vista™



Security Tokens Evolution

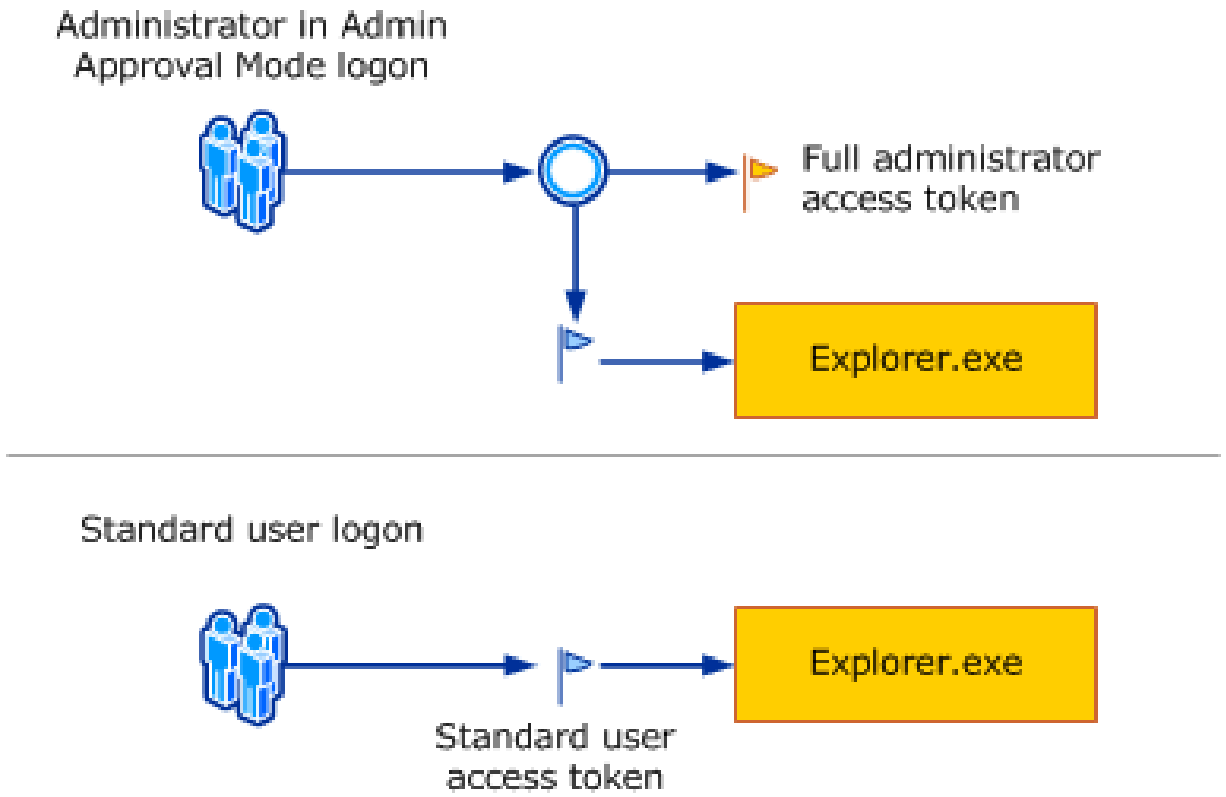
- Split Token Model

Windows Vista™



Security Tokens Evolution

- Split Token Model



Windows Vista™



Security Tokens Evolution

- Split Token Model

- Elevation Prompt

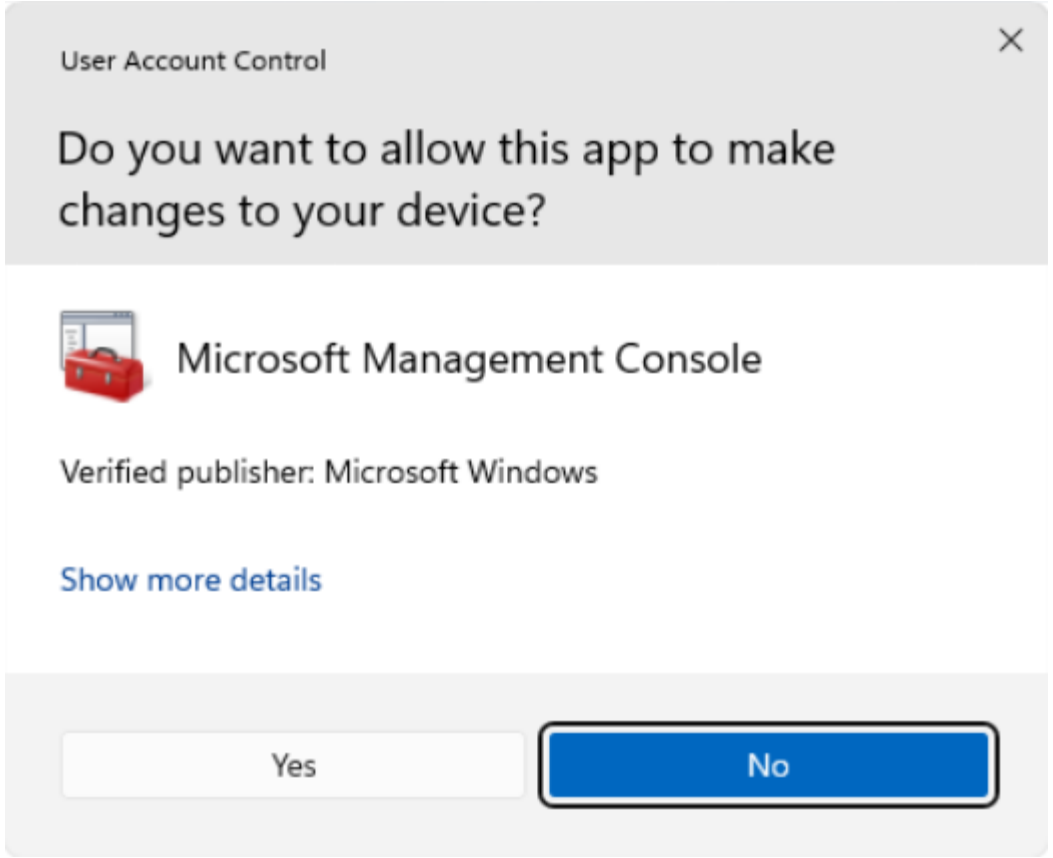
Windows Vista™



Security Tokens Evolution

- Split Token Model

- Elevation Prompt



Windows Vista™

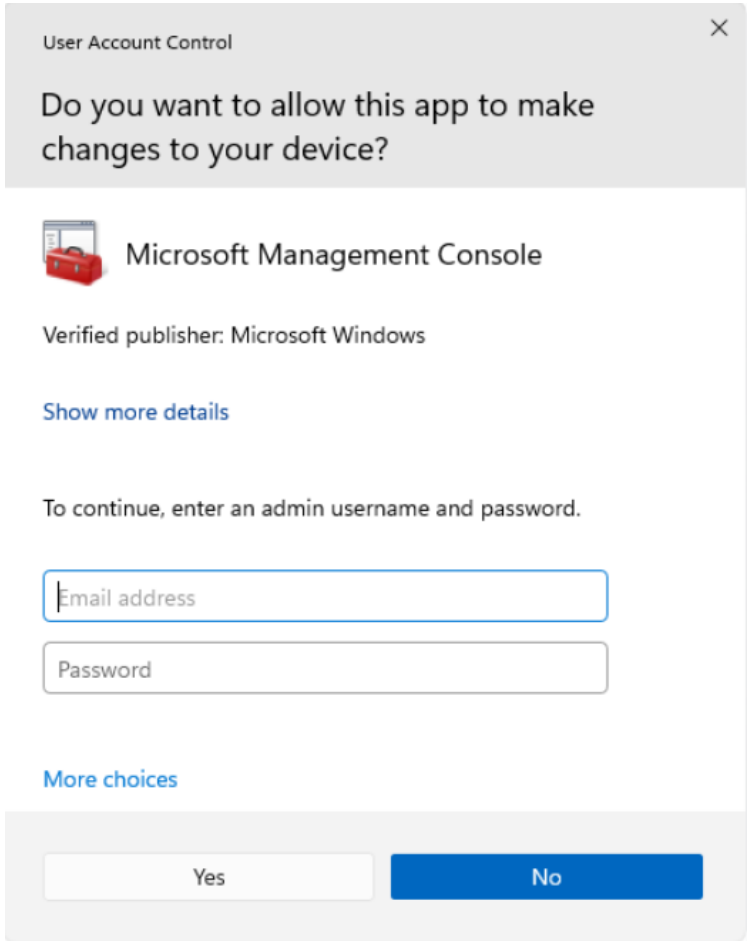


Security Tokens Evolution

- Split Token Model

- Elevation Prompt

Windows app or signed and verified app



Windows Vista™

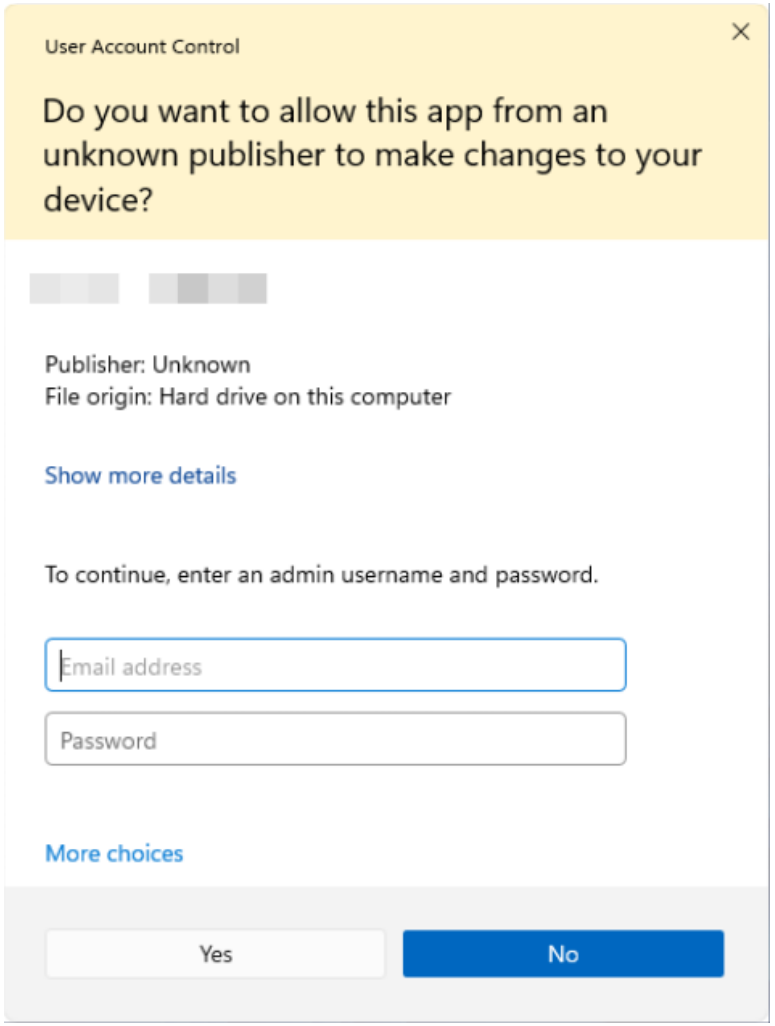


Security Tokens Evolution

- Split Token Model

- Elevation Prompt

Unsigned, or signed but not verified (trusted)



Windows Vista™

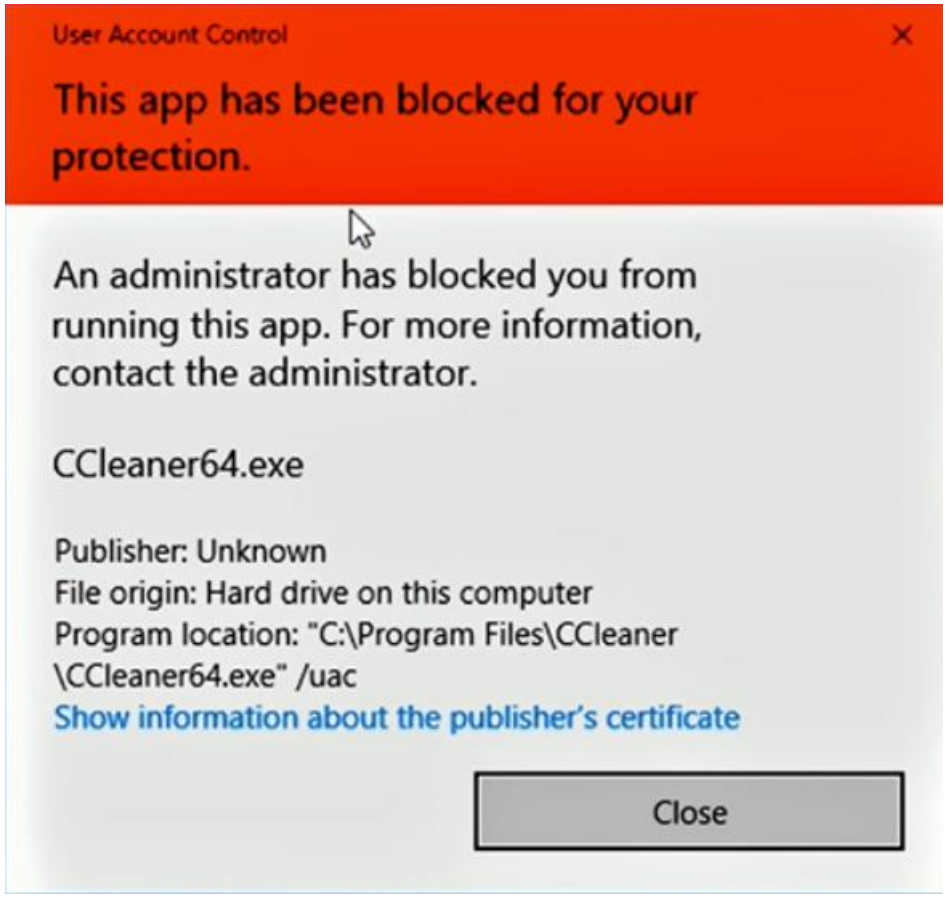


Security Tokens Evolution

- Split Token Model

- Elevation Prompt

Signed with a revoked certificate



Windows Vista™



Security Tokens Evolution

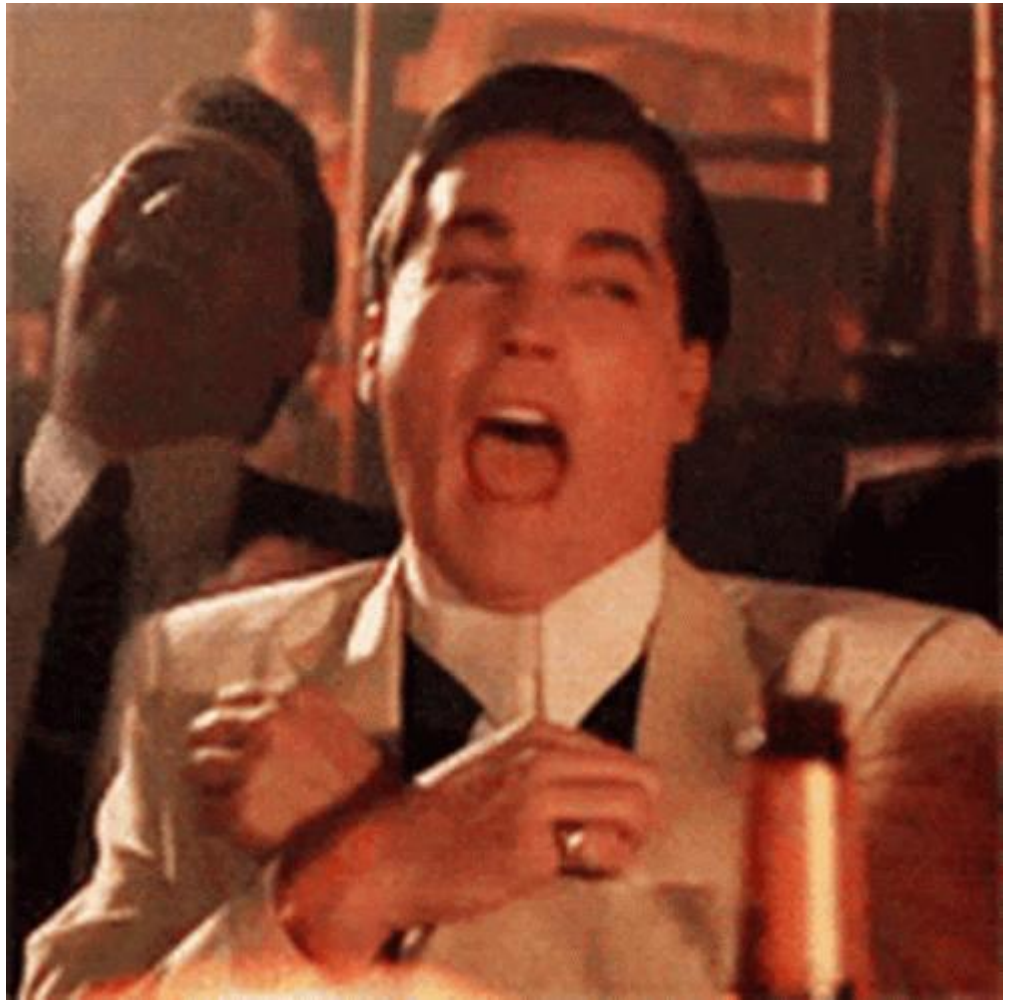
- Split Token Model
- Elevation Prompt
- Security Boundary

Windows Vista™



Security Tokens Evolution

- Split Token Model
- Elevation Prompt
- Security Boundary ?



Windows Vista™



Trivia time

What is the big problem when Microsoft doesn’t consider a technology a Security Boundary?



They don’t fix security vulnerabilities



UACMe

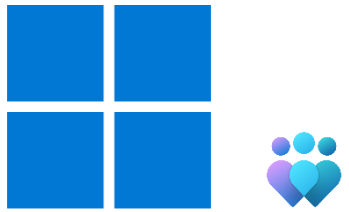
UACMe

Defeating Windows User Account Control by abusing built-in Windows AutoElevate backdoor. This project demonstrates various UAC bypass techniques and serves as an educational resource for understanding Windows security mechanisms.

> ⚠️ **Warning:** This tool demonstrates security vulnerabilities that could be exploited maliciously. Use responsibly and only in controlled environments.



Security Tokens Evolution





Security Tokens Evolution

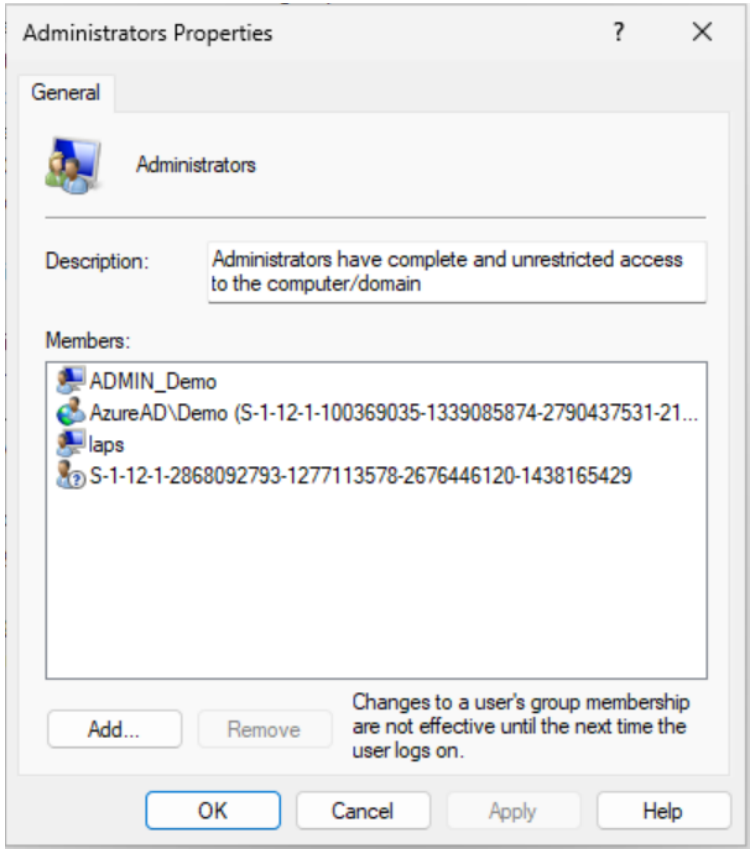
- Just-in-time elevation (SMAA)

Windows 11



Security Tokens Evolution

- Just-in-time elevation (SMAA)

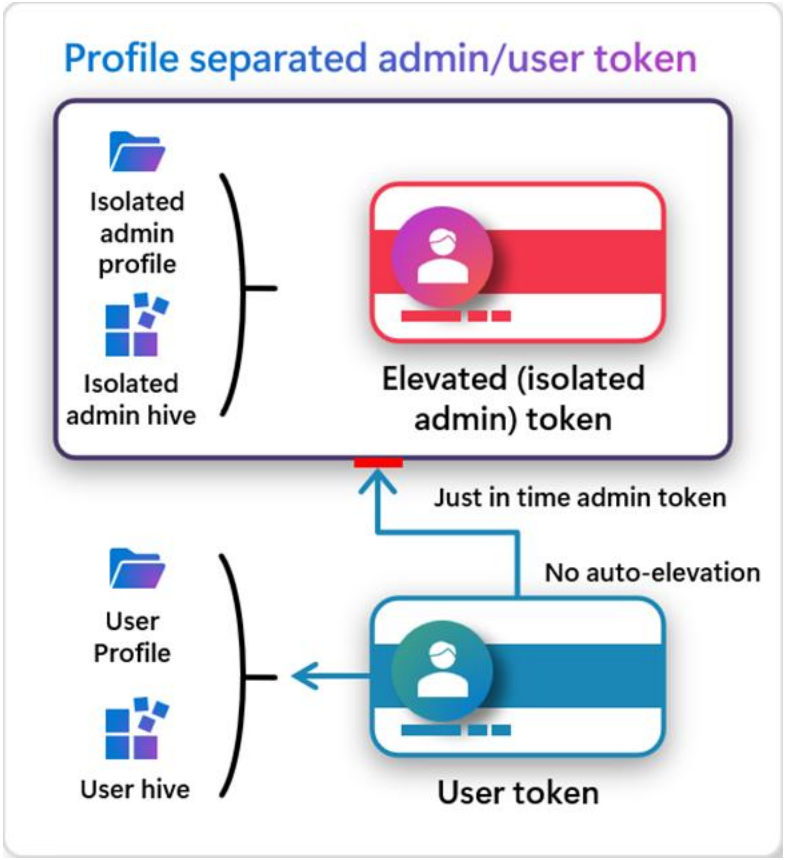


Windows 11



Security Tokens Evolution

- Just-in-time elevation (SMAA)



Windows 11



Security Tokens Evolution

- Just-in-time elevation (SMAA)

- Elevation Authentication

Windows 11



Security Tokens Evolution

- Just-in-time elevation (SMAA)

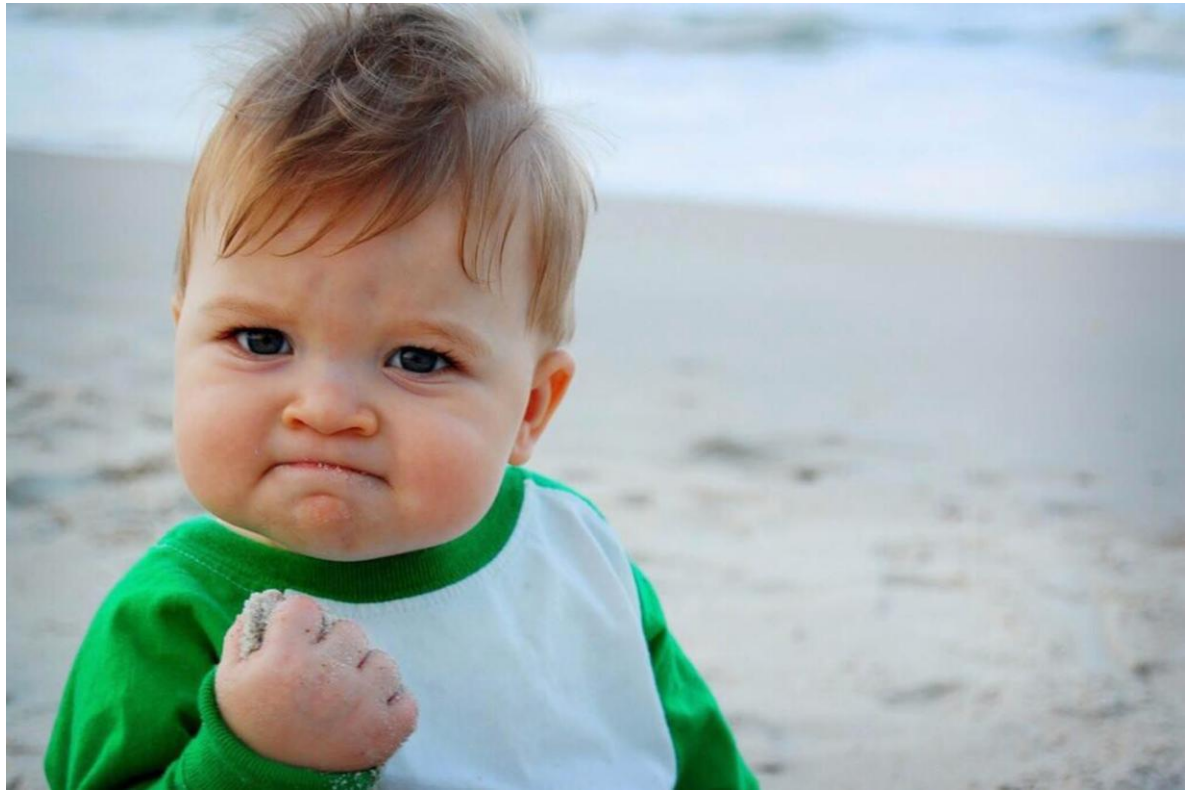
- Elevation Authentication

- Security Boundary?

Windows 11



Security Tokens Evolution



- Just-in-time elevation (SMAA)
- Elevation Authentication
- Security Boundary? YAY

Windows 11



Security Tokens Evolution

Of the 79 known UAC bypasses tested, all but one are now fully or partially mitigated. The remaining open issue around token manipulation attacks has been assigned MSRC cases and will be addressed.

Windows 11



Key Design Highlights

- New security boundary with separate profile
- Just-in-time admin token
- Removal of auto-elevation
- Windows Hello integration



Evolution differences

	Pre UAC	UAC	Admin Protection
Security Token	One Token	Split Token – Same Account	Split Token and Account
Elevation	Always	Prompt	Prompt / Authentication
Security Boundary	😍 😍 😍	No	Yes

Endpoint Privilege Management





With EPM your organization’s users can run as a standard user (without administrator rights) and complete tasks that require elevated privileges



Elevation Settings Policy

1 Configuration settings 2 Review + save

Privilege management elevation client settings

Elevation settings establish the default behaviors for the endpoint elevation client.

Endpoint Privilege Management — Enabled

Default elevation response — Deny all requests

Send elevation data for reporting *

Reporting scope *

- Deny all requests
- Require user confirmation
- Require support approval
- Not configured



Elevation Rules Policy

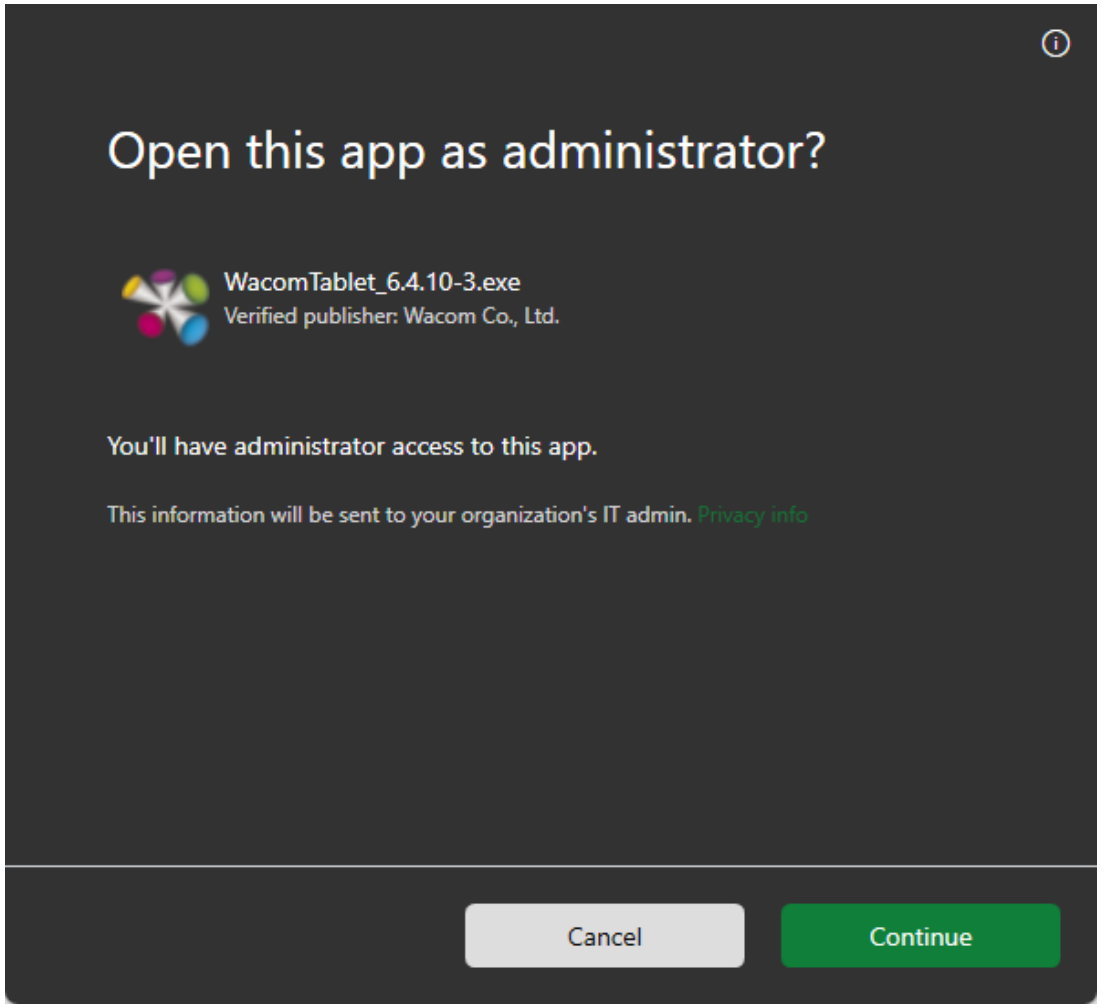
Privilege Management

Elevation Rules set the conditions for allowing users to get just-in-time access to apps and files on their devices.

Signature source	Use a certificate file in reusable settings
Certificate	
Certificate type	Publisher
Elevation type	User confirmed
Validation	Not configured
Child process behavior	Allow all child processes to run elevated
Rule name	Wacom Setup
Product name	Wacom Tablet Driver
Certificate	



Executing app





Virtual Accounts

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Task Manager

Type a name, publisher, or PID to search

Details

Run new task | End task

Name	PID	Status	User name	CPU	Memory (a...	Architecture	Platform	Command line
WacomTablet_6.4.10-3.exe	49104	Running	AzureAD_JoseSchenardie_$	00	4,240 K	x86	32 bit	"C:\Users\JoseSchenardie\Downloads\WacomTablet_6.4.10-3.exe"
Setup.exe	39180	Running	AzureAD_JoseSchenardie_$	00	6,008 K	x86	32 bit	.\setup.exe
PowerMgr.exe	8612	Running	JoseSchenardie	00	2,476 K	x86	32 bit	"C:\Windows\SysWOW64\Lenovo\PowerMgr\PowerMgr.exe"
Microsoft.Management.Servi...	4192	Running	SYSTEM	00	43,236 K	x86	32 bit	"C:\Program Files (x86)\Microsoft Intune Management Extension\Microsoft.Management.Services
dllhost.exe	13460	Running	JoseSchenardie	00	1,232 K	x86	32 bit	"C:\Windows\SysWOW64\DllHost.exe" /Processid:{8AB635F8-9A67-4698-AB99-784AD929F3B4}
armsvc.exe	30244	Running	SYSTEM	00	772 K	x86	32 bit	"C:\Program Files (x86)\Common Files\Adobe\ARM\1.0\armsvc.exe"
SystemNotificationPlugin.exe	10964	Running	SYSTEM	00	3,244 K	x64	64 bit	"C:\ProgramData\Lenovo\Udc\Hosts\x64\SystemNotificationPlugin.exe" C:\ProgramData\Lenovo
snipaste.exe	53448	Running	JoseSchenardie	00	43,000 K	x64	64 bit	"C:\Program Files\WindowsApps\45479liulios.17062D84F7C46_2.10.800.0_x64__p7pnf6hceqser\Snip
				00	8,972 K	x64	64 bit	"C:\Program Files\Common Files\Microsoft Shared\ClickToRun\OfficeClickToRun.exe" /service
ms-teams.exe	22232	Running	JoseSchenardie	00	30,208 K	x64	64 bit	"C:\Program Files\WindowsApps\MSTeams_25212.2204.3869.2204_x64__8wekyb3d8bbwe\ms-tean
				00	25,276 K	x64	64 bit	"C:\Program Files\WindowsApps\MSTeams_25212.2204.3869.2204_x64__8wekyb3d8bbwe\ms-tean
MessagingPlugin.exe	9576	Running	JoseSchenardie	00	428 K	x64	64 bit	"C:\ProgramData\Lenovo\Udc\Hosts\x64\MessagingPlugin.exe" C:\ProgramData\Lenovo\Udc\Br
LogiPresentation.exe	18552	Running	JoseSchenardie	00	176 K	x64	64 bit	"C:\Program Files\Logitech\LogiPresentation\LogiPresentation.exe"



Issues of using virtual account

- Access to user profile data
- Access to user registry hive
- User mappings
- Audits and events
- User licensed based software



What is coming

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Microsoft Intune: Intune Suite - Endpoint Privilege Management Virtual Account Support

IN DEVELOPMENT

ROLLOUT START
November 2025

Microsoft Intune

EPM will introduce a per-rule option to run elevated apps with the original user's identity context, preserving app data, save locations, and preferences. This offers a more personalized app runtime experience compared to the default virtual account.

Roadmap ID	Cloud instances(s)	Platform(s)
499155	Worldwide (Standard Multi-Tenant), GCC, GCC High, DoD	Desktop

Release phases(s)
General Availability

Added to roadmap: 07/25/2025 | **Last modified:** 07/25/2025

Microsoft 365 Roadmap

LAPS





Definition

Windows feature that automatically manages and backs up the password of a local administrator account on your Microsoft Entra-joined or Windows Server Active Directory-joined devices



Enable LAPS

Enable Microsoft Entra Local Administrator Password Solution (LAPS)

Yes | No



Create Policy

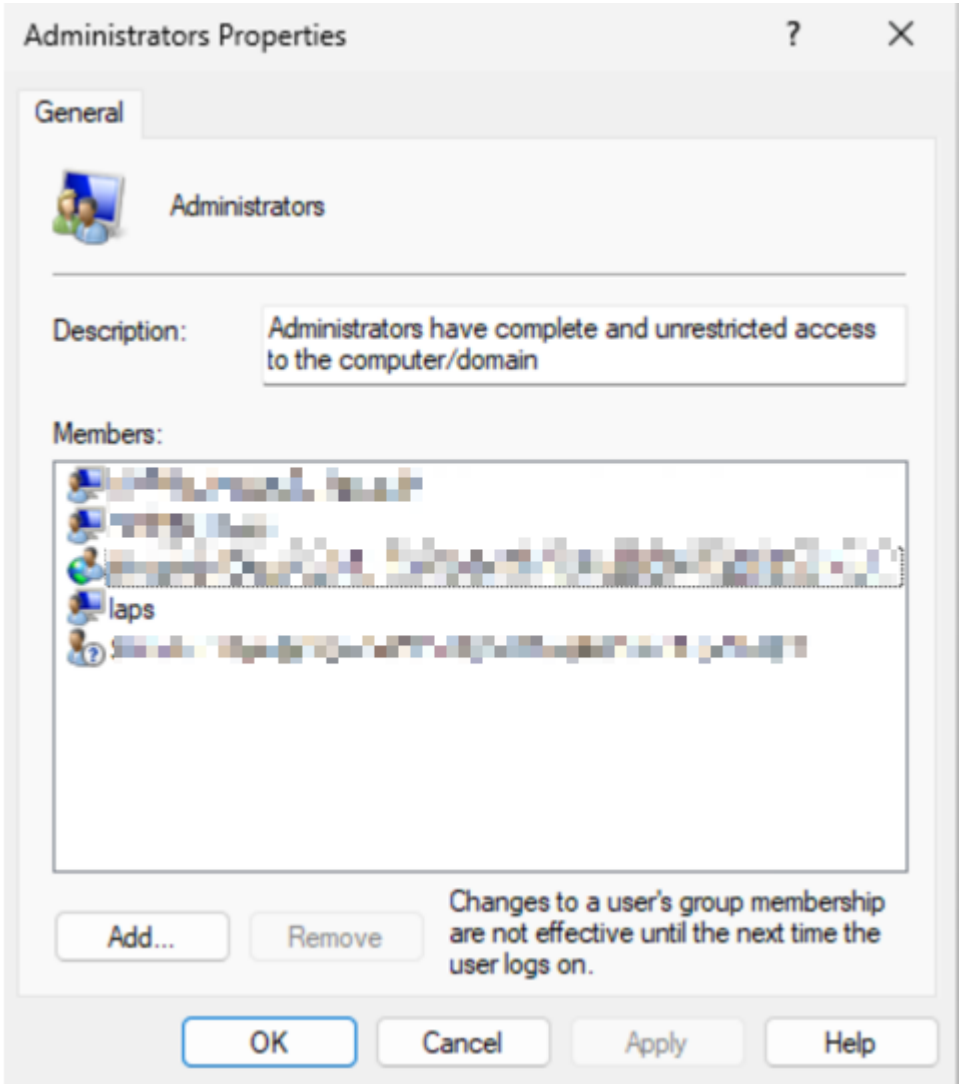
Configuration settings Edit

LAPS

Backup Directory	Backup the password to Azure AD only
Password Age Days	30
Password Complexity	Passphrase (short words with unique prefixes)
Passphrase Length	3
Password Length	14
Post Authentication Actions	Reset the password and logoff the managed account: upon expiry of the grace period, the managed account password will be reset and any interactive logon sessions using the managed account will terminated.
Post Authentication Reset Delay	24
Automatic Account Management Enabled	The target account will be automatically managed
Automatic Account Management Randomize Name	The name of the target account will not use a random numeric suffix.
Automatic Account Management Name Or Prefix	laps
Automatic Account Management Enable Account	The target account will be enabled
Automatic Account Management Target	Manage the built-in administrator account

Endpoint security - Microsoft Intune admin center




Administrators Properties
General
Administrators
Description:
Administrators have complete and unrestricted access to the computer/domain
Members:
laps
Add...
Remove
Changes to a user's group membership are not effective until the next time the user logs on.
OK
Cancel
Apply
Help



Licenses, Accounts, etc

	UAC	Administrator Protection	LAPS	Endpoint Privilege Management
License	Part of Windows	Part of Windows	Part of any Windows	Intune Suite / Stand Alone
Account	Self	SMAA ADMIN_%name%	Local account (protected)	Virtual account %name%_$
Elevation	Broad	Broad	Broad	Granular

Usage Scenarios





When to use

User Account Control

Administrator Protection

EPM

LAPS

Intune



Resources

- Dual enrolment
- Evolving the Windows User Model – A Look to the Past
- Evolving the Windows User Model – Introducing Administrator Protection
- Enhance your application security with administrator protection
- How to protect your administrator users on the device
- Administrator Protection on Windows 11
- hfiref0x/UACME: Defeating Windows User Account Control



Thank You

